

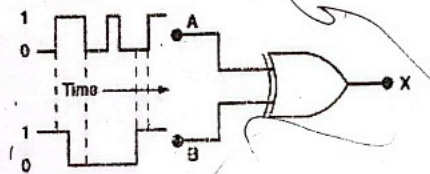
- ① Answer any 05 out of 06 Questions (Split answers are highly discouraged)
- [A.] Simplify $F(w, x, y, z) = \Sigma(1, 3, 7, 11, 15)$ which has the don't-care conditions $d(w, x, y, z) = \Sigma(0, 2, 5)$ and show that an expression with the minimum number of literals is not necessarily unique. 7
- [B.] i. Find the complement of the function $F_1 = xyz' + x'y'z'$ and $F_2 = x(yz' + yz)$ applying De Morgan's theorem. 7
- ii. Prove that $f_1 = m_1 + m_2 + m_6 + m_7 = M_0, M_3, M_4, M_5$.

- ② [A.] Consider $F(A, B, C, D) = \Sigma(0, 2, 3, 5, 7, 8, 9, 10, 11, 13, 15)$. Find out and show prime implicants and essential prime implicants by the Karnaugh map from the above expression. 7

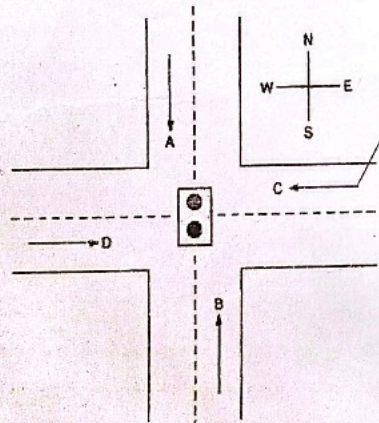
- [B.] Simplify the following Boolean function by using the tabulation method: $F = \Sigma(0, 1, 2, 8, 10, 11, 14, 15)$. 7

- 3 [A.] A four-bit binary number is represented as $A_3A_2A_1A_0$, where A_3, A_2, A_1 , and A_0 represent the individual bits and A_0 is equal to the LSB. Design a logic circuit that will produce a HIGH output whenever the binary number is greater than 0010 and less than 1000. 2

- [B.] i. Determine the output waveform for the circuit of figure. 3
- ii. Repeat with the B input held LOW.
- iii. Repeat with B held HIGH.



[C.]



4 8 12 13 14
E-W = 0

The above figure shows the intersection of a main highway with a secondary access road. Vehicle-detection sensors are placed along lanes C and D (main road) and lanes A and B (access road). These sensor outputs are LOW (0) when no vehicle is present and HIGH (1) when a vehicle is present. The intersection traffic light is to be controlled according to the following logic:

- The east-west (E-W) traffic light will be green whenever *both* lanes C and D are occupied.
- The E-W light will be green whenever *either* C or D is occupied but lanes A and B are not *both* occupied.
- The north-south (N-S) light will be green whenever *both* lanes A and B are occupied but C and D are not *both* occupied.
- The N-S light will also be green when *either* A or B is occupied while C and D are *both* vacant.
- The E-W light will be green when *no* vehicles are present.

Using the sensor outputs A, B, C, and D as inputs, design a logic circuit to control the traffic light. There should be two outputs, N-S and E-W, that go HIGH when the corresponding light is to be green. Simplify the circuit as much as possible and show *all* steps.

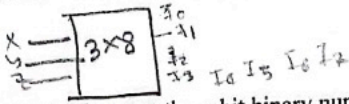
- [D.] Implement following Boolean function using multilevel NAND gates.

$$F = A(B + CD) + BC'$$

4. [A.] Define universal gate. Implement a combinational circuit which convert four bit excess 3 code to four bit BCD code.

[B.] Define adder. Implement a full-adder circuit with a decoder and two OR gates. From the truth table of the full-adder (Section 4-3), we obtain the functions for this combinational circuit in sum of minterms: $S(x, y, z) = \sum(1, 2, 4, 7)$

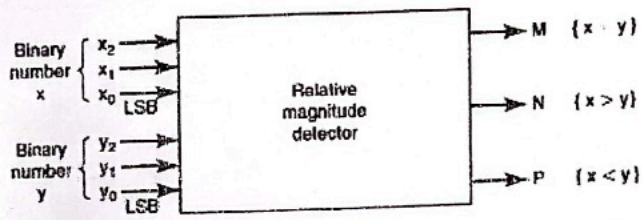
$C(x, y, z) = \sum(3, 5, 6, 7)$



[C.] Figure represents a relative-magnitude detector that takes two three-bit binary numbers, $x_2x_1x_0$ and $y_2y_1y_0$, and determines whether they are equal and, if not, which one is larger. There are three outputs, defined as follows:

- i. $M = 1$ only if the two input numbers are equal.
- ii. $N = 1$ only if $x_2x_1x_0$ is greater than $y_2y_1y_0$.
- iii. $P = 1$ only if $y_2y_1y_0$ is greater than $x_2x_1x_0$.

Design the logic circuitry for this detector. The circuit has six inputs and three outputs and is therefore much too complex to handle using the truth table approach.



$M = x_2x_1x_0 = y_2y_1y_0$
 $N = x_2\bar{y}_2 + \bar{x}_2y_2$
 $P = y_2\bar{x}_2 + \bar{y}_2x_2$

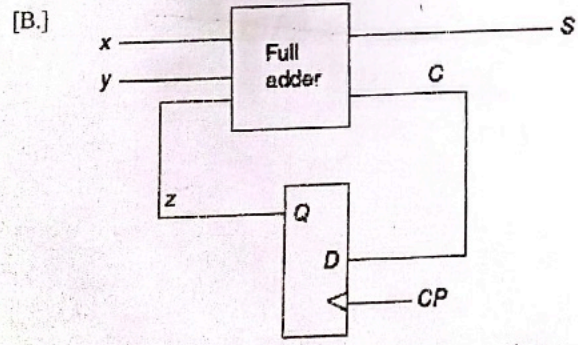
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[D.] Define programmable read-only memory. Implement the following function with a multiplexer and demultiplexers and show that the demultiplexure delivers which was exactly combined by multiplexure.

$F(A, B, C) = \sum(1, 3, 5, 6)$

3 variable
(3-1) = 2 selection line
 $2^2 = 4$ Input
 I_0, I_1, I_2, I_3

5. [A.] What is flip-flop? Design a D flip-flop.



The full-adder of figure receives two external inputs x and y ; the third input z comes from the output of a D flip-flop. The carry output is transferred to the flip-flop every clock pulse. The external S output gives the sum of x , y , and z . Obtain the state table and state diagram of the sequential circuit.

[C.] Differentiate between sequential circuit and combinational circuit. What is the problem found in RS flip-flop? Explain how it is solved in JK flip flop.

[D.] Bring out the differences between edge triggered and level triggered flip-flop. What is the necessity of master-slave flip-flop? Explain working of D master-slave flip-flop. Realize with all NOR gates.

6. [A.] Define registers. Design a 4-bit register with parallel load using D flip-flops.

[B.] What is bidirectional shift register? The content of a 4-bit shift register is initially 1101. The register is shifted six times to the left, with the serial input being 101101. What is the content of the register after each shift?

[C.] Define binary ripple counter. Determine and show the count sequence for a binary ripple counter.

[D.] Design the basic memory unit architecture showing communication with environment. Show the construction of a Johnson counter.

Dept. of Computer and Communication Engineering
Faculty of Computer Science and Engineering
Patuakhali Science and Technology University
 Dumki, Patuakhali-8602, Bangladesh

Final Examination of B. Sc. Engineering in CSE Level: 2 Semester: II Session: 2019-2020

Course Code
CCE-221

Course Title
Digital Logic Design

July-December
2021

Credit: 03
Time: 03 Hr
Marks: 70

Answer any 05 out of 06 Questions (Split answers are highly discouraged)

1. [A.] Simplify the Boolean function $F = A'B'C'D + B'CD' + A'BCD' + AB'C'$ and implement the resultant using logic gates. 7
- [B.] a) Find the complement of the function $F_1 = xyz' + x'y'z'$ and $F_2 = x(yz' + yz)$ applying De Morgan's theorem. 7
 b) Prove that $f_1 = m_0 + m_1 + m_6 + m_7 = M_2 M_3 M_4 M_5$.
2. [A.] Simplify the Boolean function $F(w, x, y, z) = \sum(1, 3, 7, 11, 15)$ which has the don't-care conditions $d(w, x, y, z) = \sum(0, 2, 5)$ show "An expression with the minimum number of literals is not necessarily unique". 7
- [B.] Simplify the Boolean function $f(A, B, C, D) = \sum(0, 1, 2, 3, 5, 7, 8, 10, 12, 13, 15)$ using the tabulation method. 7
3. [A.] It is necessary to multiply two binary numbers, each two bits long, in order to form their product in binary. Let the two numbers be represented by a_1, a_0 and b_1, b_0 , where subscript 0 denotes the least significant bit. 2
 - i. Determine the number of output lines required.
 - ii. Find the simplified Boolean expressions for each output.
- [B.] Define universal gate. Design a 4-bit parity checker. 3
- [C.] Design a BCD-to-seven-segment decoder in a combinational circuit that accepts a decimal digit in BCD and generates the appropriate outputs for selection of segments in a display indicator used for displaying the decimal digit. The seven outputs of the decoder (a, b, c, d, e, f, g) select 7

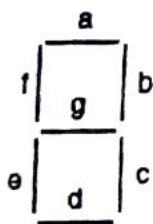


Fig. (a) Segment designation

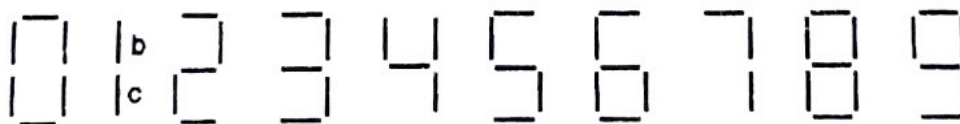


Fig. (b) Numerical designation for display

the corresponding segments in the display as shown in Fig. (a). The numeric designation chosen to represent the decimal digit is shown in Fig. (b). Design the BCD-to-seven-segment decoder circuit.

- [D.] Implement following Boolean function using exclusive or and AND gates 2
- $$F = ABCD + A'B'C'D' + ABC'D' + A'B'CD$$

4. ☒ [A.] What is the disadvantage of 4-bit full adder? Design a 4-bit full-adder with look-ahead carry. 5
☒ [B.] Define encoder. Design Octal-to-binary encoder. 3
☒ [C.] What is multiplexers? Design a 4-bit magnitude comparator. 4
☒ [D.] Write down the purpose of EPROM. Implement the following function with a multiplexer: 2

$$F(A, B, C, D) = \Sigma(0, 1, 3, 4, 8, 9, 15)$$
5. ☒ [A.] Design a 4-line to 2-line priority encoder. Include an output E to indicate that at least one input is a 1. 3
☒ [B.] How many don't-care inputs are there in a BCD adder? Design a 4-bit BCD adder. 4
☒ [C.] What are the difference between latch and flip-flop? Design a T-flip-flop and determine its charactersitics equation. 4
☒ [D.] Consider a JK' flip-flop, i.e., a JK flip-flop with an inverter between external input K' and internal input K. 3
 i. Obtain the flip-flop characteristic table. chapter 6
 ii. Obtain the characteristic equation.
 iii. Show that tying the two external inputs together forms a D flip-flop.
6. [A.] The content of a 4-bit shift register is initially 1101. The register is shifted six times to the right, with the serial input being 101101. What is the content of the register after each shift? chapter 7 3
 [B.] What is the difference between serial and parallel transfer? Design a serial adder using a sequential-logic procedure. chapter 7 4
 [C.] Define binary ripple counter. Determine and show the count sequence for a binary ripple counter. 3
 [D.] Define memory unit architecture. Show the construction of a Johnson counter. 4

Dept. of Computer and Communication Engineering
Faculty of Computer Science and Engineering
Patuakhali Science and Technology University
Dumki, Patuakhali-8602, Bangladesh

F-removal Examination of B. Sc. Engineering in CSE Level: 2 Semester: II Session: 2019-2020

Course Code
CCE 221

Course Title
Digital Logic Design

Credit: 03
Time: 03 Hrs.
Marks: 70

Answer any 05 out of 06 Questions (Split answers are highly discouraged)

1. ~~A~~ "An expression with the minimum number of literals is not necessarily unique"-Illustrate the significance of the above statement with appropriate example. 07
1. B 07
1. ~~I.~~ Convert $(5114)_{10}$ to base-5 number.
1. ~~II.~~ Prove that $f_1 = m_2 + m_1 + m_5 + m_7 = M_0, M_3, M_4, M_6$.
2. ~~A~~ 1. What are the advantages and disadvantages of digital techniques over analog technique? 07
2. ~~II.~~ Simplify the Boolean function $f(w, x, y, z) = \sum(0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 14, 15)$.
2. ~~B~~ Simplify the Boolean function $F = A'B'C' + B'CD' + A'BCD' + AB'C'$ and implement the resultant using logic gates. 07
3. ~~A~~ 1. Define combinational circuit. Design a combinational circuit that accepts a three-bit binary number and generates an output number equal to the square of the input number. $B_1 = A_1$ $B_2 = B_0$ $B_3 = A_2 \bar{A}_1$ $B_4 = A_2(A_2 \oplus A_3)$ $B_5 = A_3(A_1 + \bar{A}_2)$ $B_6 = A_3 A_2$ 07
2. ~~II.~~ Design a full adder using half adder with mathematical backgrounds.
3. ~~B~~ Design and describe the mechanism of a magnitude comparator circuit for 4-bit binary numbers $A = A_3 A_2 A_1 A_0$ and $B = B_3 B_2 B_1 B_0$. The outputs are F, G, and H, I, J, K where F is 1 if $A > B$, G is 1 if $A = B$, H is 1 if $A < B$, I is 1 if $A \geq B$, J is 1 if $A \leq B$, K is 1 if $A \neq B$. 07
4. ~~A~~ Define decoder. Design and describe the mechanism of a BCD adder circuit. 07
4. ~~B~~ Why multiplexer is needed? Design and describe the mechanism of a 4-bit multiplexing and demultiplexing technology with synchronizing control. 07
5. ~~A~~ i. Define Synchronous and Asynchronous sequential circuits with example. 07
ii. Design a JK flip-flop with its characteristics table and equation. page-208
5. ~~B~~ Define register. Design a 4-bit register with parallel load using D flip-flop. 07
SP n n
6. ~~A~~ 1. Design a serial adder using a sequential logic procedure. 07
ii. Design a 4-bit binary ripple counter using flip-flops that trigger on the positive edge.
6. ~~B~~ What is the difference between serial and parallel transfer? The content of a 4-bit shift register is initially 1101. The register is shifted eight times to the right, with the serial input being 101101. What is the content of the register after each shift? 07

Patuakhali Science and Technology University

4th Semester (Level-2, Semester-II), Final Examination of B.Sc. Engg. (CSE), July-December/2019,

Session: 2017-18

Course Code: CCE-221 Course Title: Digital Logic Design

Credit Hour: 3.00 Full Marks: 70 Duration: 3 Hours

[Figures in the right margin indicate full marks. Split answering of any question is not recommended]

Answer any 5 of the following questions.

1. a. Simplify the Boolean function $F = A'B'C'D + B'CD' + A'BCD' + AB'C'$ and implement the resultant using logic gates. 07
- b. "An expression with the minimum number of literals is not necessarily unique"-Illustrate the significance of the above statement with appropriate example. 07
2. a. Simplify the Boolean function $f(w,x,y,z) = \sum(0,1,2,3,4,5,6,8,9,10,11,12,13,15)$. 07
- b. Find the complement of the function $F_1 = xyz' + x'y'z'$ and $F_2 = x(yz' + yz)$ applying De Morgan's theorem. 04
- c. Prove that $f_1 = m_0 + m_1 + m_6 + m_7 = M_2 M_3 M_4 M_5$. 03
3. a. Design a BCD-to Excess-5 code converter. The inputs of BCD are A, B, C, D and outputs of excess-5 are w, x, y, z and use only basic gates for implementation. 4
- b. Write down the differences between combinational and sequential circuit. Differentiate between synchronous and asynchronous counter. Which is easier to design? 5
- c. Draw block diagram of BCD adder that is composed of two 4-bit binary adder and external gates. Explain how it manages the BCD addition. 5
4. a. Define decoder. Design a 3-input circuit that will output '1' when the majority of the inputs are 1. 4
- b. Suppose you have some 2 to 4 line decoder. But you need 4 to 16 line decoder. How can you built 4 to 16 line decoder using your decoders? 5
- c. Design a full-adder circuit with a decoder and two OR Gates. Functions for the sum and carry are as followings : 5

$$\text{Sum: } S(x, y, z) = \sum(1, 2, 4, 7)$$

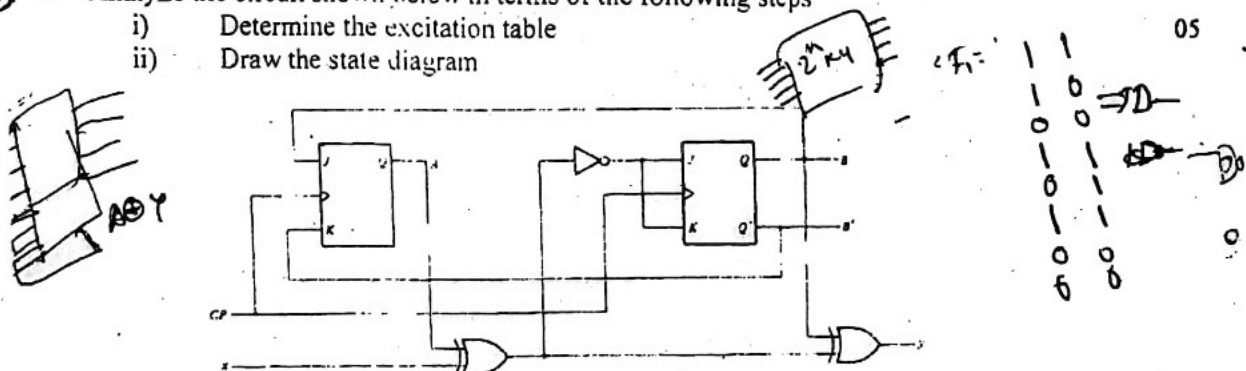
$$\text{Carry: } C(x, y, z) = \sum(3, 5, 6, 7)$$

- 4
5. a. Implement the following boolean function using 4 to 1 multiplexer
 $F = \sum(1, 3, 5, 6)$
- b. Suppose you have a bit sequence '10011' in register A. Shift the bits to register B and C sequentially. Design the circuit and show the shift sequences.
- 5
- c. Explain different types of clocked circuit. Draw truth table and block diagram of 4-bit binary parallelmode-10 counter using JK flip-flop.
- 5
6. a. Design a sequential circuit with two D flip-flops A and B, and one input x_{in} .
When $x_{in} = 0$, the state of the circuit remains the same. When $x_{in} = 1$, the circuit goes through the state transitions from 00 to 11, to 10, to 01, back to 00, and repeats.
- 4
- b. Design a 4-bit parallel load register using a 4-bit simple register, which outputs the stored data in negative edge of clock signal.
- 5
- c. Draw truth table and block diagram of 4-bit binary ripple counter using JK flip-flop.
- 5

[Figures in the right margin indicate full marks. Split answering of any question is not recommended.]

Answer any 5 of the following questions.

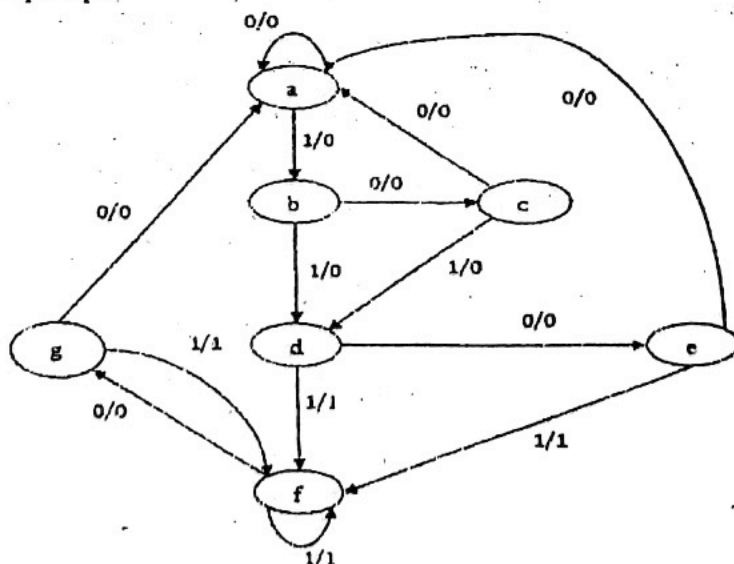
1. a. What do you mean by complements? Make a comparison between 1's and 2's complements. 02
- b. Convert octal 623.77 to decimal, binary and hexadecimal. 03
- c. Represent the decimal number 8620 (i) in biquinary (ii) in excess-3 code (iii) in 2,4,2,1 code. 03
- d. Draw switching circuits of two input NAND, NOR and AND operation. 03
- e. Express the following function in a product of maxterm form. $F(w,x,y,z) = y'z + wxz' + w'x'z$ 03
2. a. What is duality principle? Prove $x(x+y) = x$ by duality. 02
- b. Simplify the Boolean function F using the don't care conditions d. 03
- c. Simplify the following Boolean functions by means of the tabulation method. 05
- d. Draw and describe 4-bit Bidirectional shift register with parallel load. 04
3. a. What is the difference between serial and parallel transfer? What type of register is used in each case? 02
- b. Define combinational logic circuit. Show that a full-subtractor can be constructed with two half-subtractors and an OR gate. 03
- c. Design combinational circuit for even parity generation and even parity checker. 04
- d. Implement the function $F = AB + CD + E$ using NAND gate only. 03
4. a. How can you determine the number of links in PLA? 02
- b. Design of a 4 bit magnitude comparator circuit. 05
- c. Implement a full adder circuit with a decoder and two OR gates. 03
- d. What is the basic difference between flip flop and latch? Explain the operation of toggle flip flop with relevant logic diagram and truth table. 06
5. a. A combinational circuit is defined by the following functions: 04
- b. Design a BCD counter with JK flip flop. 05
- c. What do you mean by ROM? Draw the logic construction of a 32×4 ROM. Also design a BCD to excess-3 code converter using ROM. 05
6. a. Analyze the circuit shown below in terms of the following steps 05



- b. What do you mean by positive and negative logic? Show that a positive logic AND gate is equivalent to a negative logic OR gate? 03
- c. Draw the logic diagram of clocked RS flip flop with four NAND gates 02
- d. Design a counter with the following sequence: 0, 4, 2, 1, 6 and repeat. Use D flip flop to design it. 04

[Figures in the right margin indicate full marks. Split answering of any question is not recommended.]
 Answer any 5 of the following questions.

- [1] a. Obtain 1's complement and 2's complement of the following binary numbers: 1010101, 0111000, 0000001 02
- b. Convert the following numbers to decimal: (i) $(1001001.011)_2$ (ii) $(0.342)_6$ (iii) $(1032.2)_4$ 03
- c. Convert decimal digit 0 to 9 into BCD, 84-2-1, 2421 and biquinary codes. 03
- d. Perform the subtraction operation with the following binary numbers using (1) 2's complement and (2) 1's complement. (i) 10010-10011 (ii) 100-110000 (iii) 0001-1001 03
- e. Draw switching circuits of two input NAND, NOR and AND operation. 03
- [2] a. Express the Boolean function $F(x,y,z)=(xy+z)(y+xz)$ in a product of maxterm form 03
- b. Draw and describe 4-bit Bidirectional shift register with parallel load. 04
- c. What do you understand by canonical and standard forms? 02
- d. Classify digital IC gates according to logic operation. Write down some special characteristics of digital IC logic families. Also, give numerical value of each type. 03
- e. What is duality principle? Prove $x(x+y)=x$ by duality. 02
- [3] a. Design of a counter circuit that will count binary 000 to 111. 05
- b. Simplify the following Boolean function in (i) sum of products and (ii) product of sums.
 $F(A,B,C,D)=\Sigma(0,1,2,5,8,9,10)$ 03
- c. Determine the prime implicants of the function: $F(w,x,y,z)=\Sigma(1,4,6,7,8,9,10,11,15)$ 04
- d. Design of a half subtractor circuit. 02
- [4] a. Design of a combinational circuit that will convert BCD into Excess-3 codes. 05
- b. Design combinational circuit for odd parity generation and odd parity checker. 04
- c. Design of a BCD adder circuit. 05
- [5] a. Design of a magnitude comparator circuit. 05
- b. Implement the following function with a multiplexer: $F(A,B,C,D)=\Sigma(0,1,3,4,8,9,15)$ 03
- c. Define the following terms: Demultiplexers, Encoder, PLA 03
- d. What do you mean by sequential logic circuits? Draw the excitation table of RS and JK flipflops. 03
- [6] a. Define the following terms: flip-flop, Register, Memory unit 02
- b. Design of a sequential circuit for the state diagram shown in figure. Simplify the state diagram. Use the reduced state table with assignment $a=000$, $b=010$, $c=011$, $d=101$, $e=111$. Design the circuit using JK flip-flops. 12



[Figures in the right margin indicate full marks. Split answering of any question is not recommended.

Write the full question number e.g. 5(B) (II) before the answer paragraph]

Answer any 5 of the following questions

- | | | | |
|---|---|---|----|
| 1 | ✓ | Design a BCD adder circuit. | 04 |
| 1 | B | Implement the following function with a multiplexer:
$F(P,Q,R,S) = \sum(0,2,5,6,8,12,14,15)$ | 04 |
| 1 | ✓ | Design a combinational circuit using ROM which accepts a 3-bit binary number and generates an output binary number equal to the square of the input number. | 04 |
| 1 | ✗ | Write short notes on Multiplexers. | 02 |
| 2 | A | Write down the excitation table of RS and JK flip-flop. | 03 |
| 2 | B | Design a sequential circuit as shown in the fig. 2(B) by using JK flip flop. Use reduced state table and state assignment of 000, 010, 011, 101, and 111. | 09 |

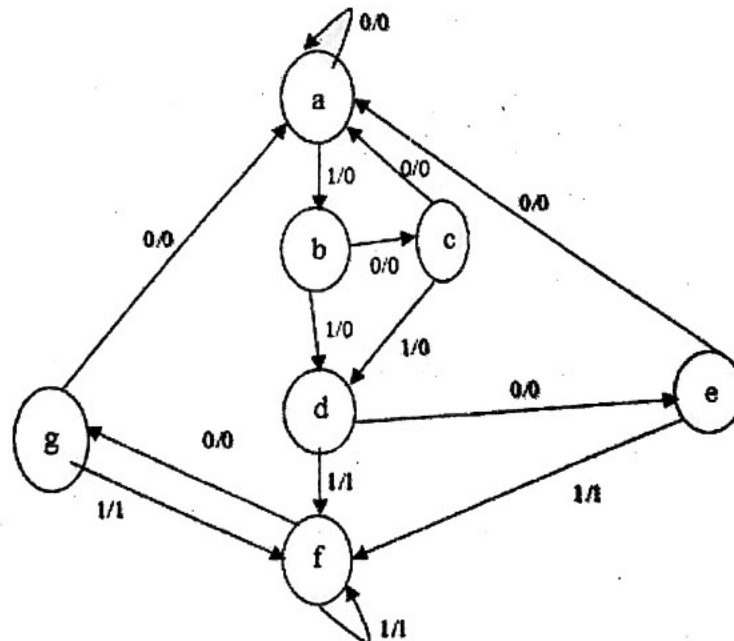


Fig: 2(B)

- | | | | |
|---|---|----------------------------------|----|
| 2 | C | Define register and memory unit. | 02 |
|---|---|----------------------------------|----|

- 3 A Design a counter that has a repeated sequence of 000,001,010,011,100,101,110. Use JK flip flop. 06
- 3 B Design a sequential circuit with JK flip-flops to satisfy the following state equations: 04
 $A(t+1) = A'B'CD + A'B'C + ACD + AC'D'$
 $B(t+1) = A'C + CD' + A'BC'$
 $C(t+1) = B'$
 $D(t+1) = D'$
- 3 C Discuss serial transfer using two 4 bit shift registers of a clocked sequential circuit. 04
- 4 A 07
~~I.~~ Implement the following equations using logic gates $F = xy'z + x'yz + xy'$ and $F = xy' + xz$.
~~II.~~ Subtract $(28532 - 345920)_{10}$ using 10's complement and $(11010010 - 10001100)_2$ using 1's complement.
- 4 B 07
~~I.~~ Design a full adder using half adder and necessary gates.
~~II.~~ Find the complement of the function $F_1 = x'yz' + x'y'z$ and $F_2 = x(y'z' + yz)$ applying De Morgan's theorem.
- 5 ~~A~~ "An expression with the minimum number of literals is not necessarily unique"-Illustrate the significance of the above statement with appropriate example. 07
- 5 B 07
~~I.~~ Convert $(511.4)_{10}$ to base-5 number.
~~II.~~ Prove that $f_1 = m_0 + m_1 + m_5 + m_7 = M_2.M_3.M_4.M_6$.
- 6 A 07
~~I.~~ What are the advantages and disadvantages of digital techniques over analog technique?
~~II.~~ Simplify the Boolean function $f(w,x,y,z) = \sum(0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15)$.
- 6 ~~B~~ Simplify the Boolean function $F = A'B'C' + B'CD' + A'BCD' + AB'C'$ and implement the resultant using logic gates. 07